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SIEM SECURITY ANALYSIS REPORT

Generated: 2026-03-26 17:43:06 | Log Source Type: MySQL Error

Total Events	5000
Parsed Successfully	5000 (100%)
Risk Level	CRITICAL
ML Detected Threats	0

Overall Risk Score: 100/100 (CRITICAL)

1. EXECUTIVE SUMMARY

This comprehensive security report presents the analysis of **5,000** log events from the SIEM system. The analysis successfully parsed **5,000** events (100% success rate) with **0** events failing to parse.

■ **No ML-detected threats** were found in the analyzed log sample. The security posture appears normal.

2. AI-POWERED THREAT ANALYSIS

This section contains detailed threat analysis generated by our AI SOC Analyst.

SIEM Security Analysis Report

Date: 2026-01-16

Analyst: AI Security Analyst

1. EXECUTIVE SUMMARY

This report highlights a concerning number of SQL injection attempts against the MySQL database, indicating a potential vulnerability in the application layer. Additionally, brute-force login attempts targeting the 'root' user from external IPs raise concerns about unauthorized access and potential data breaches. Immediate action is required to mitigate these risks.

2. KEY FINDINGS

| Finding | Severity | Description |

|---|---|---|

| SQL Injection Attempts | **CRITICAL** | Multiple high-confidence SQL injection attempts detected, potentially leading to data exfiltration, modification, or deletion. |

| Brute-Force Login Attempts (Root User) | **HIGH** | Repeated failed login attempts to the 'root' user from external IP addresses, indicating a brute-force attack. |

| Insecure PID File Configuration | **MEDIUM** | The MySQL server is configured with an insecure PID file, potentially allowing local privilege escalation. |

| Aborted Database Connections | **LOW** | Numerous aborted connections to the 'production' database may indicate resource exhaustion or network instability. |

| Communication Packet Timeout | **LOW** | Frequent timeout errors reading communication packets may indicate network latency or server overload. |

3. ATTACK CHAIN ANALYSIS

The detected activity suggests the following attack chain:

* **Reconnaissance (TA0043):** The attacker(s) may be probing the application for vulnerabilities. The "Got timeout reading communication packets" could be a side effect of this.

* **Exploitation (TA0002):** The numerous SQL injection alerts (T1190) indicate attempts to exploit vulnerabilities in the application's database interaction. The `Query '1; EXEC xp\_cmdshell('whoami')--' failed with syntax error` log entry shows an attempt to execute system commands via SQL injection, which is a clear indicator of malicious intent.

* **Credential Access (TA0006):** The failed login attempts to the 'root' user from external IPs (T1110) suggest a brute-force attack aimed at gaining unauthorized access to the database server.

This attack chain aligns with common web application attack patterns. The attacker is attempting to leverage SQL injection to gain control of the database and potentially the underlying system.

4. THREAT INTELLIGENCE

* **IOCs:**

* **IP Addresses:**

* 193.42.33.45: Source of multiple failed 'root' login attempts.

* 195.54.160.149: Source of failed 'root' login attempts.

* 45.146.165.37: Source of failed 'sa' login attempts.

* **Username:**

* root: Target of brute-force attacks.

* sa: Target of brute-force attacks.

* **SQL Injection Payloads:**

* `1; EXEC xp\_cmdshell('whoami')--`: Attempt to execute system commands.

* **Suspicious IPs:** The IP addresses listed above should be investigated further and potentially blocked at the firewall. Consider using threat intelligence feeds to enrich these IPs with reputation data.

* **Attacker TTPs:**

* **SQL Injection (T1190):** Exploiting vulnerabilities in the application's database interaction.

* **Brute Force (T1110):** Attempting to guess credentials through repeated login attempts.

* **Command Execution (T1059):** Attempting to execute system commands via SQL injection.

5. RISK ASSESSMENT

Current Risk Level: HIGH

Justification: The detection of active SQL injection attempts poses a significant risk to the confidentiality, integrity, and availability of the database. Successful exploitation could lead to data breaches, system compromise, and reputational damage. The brute-force login attempts further exacerbate the risk by potentially granting unauthorized access to the database server. While the aborted connections and timeout errors are less critical, they can impact

system performance and availability. The insecure PID file configuration presents a local privilege escalation vulnerability that could be exploited by an attacker who has already gained access to the system.

6. RECOMMENDATIONS

1. **Patch SQL Injection Vulnerabilities (Immediate):** Identify and patch the SQL injection vulnerabilities in the application code. Implement parameterized queries or prepared statements to prevent SQL injection attacks.
2. **Implement Web Application Firewall (WAF) (Immediate):** Deploy a WAF to filter malicious traffic and block SQL injection attempts. Configure the WAF with rules to detect and prevent common SQL injection patterns.
3. **Enforce Strong Password Policies (High):** Enforce strong password policies for all database users, including the 'root' user. Require complex passwords and regular password changes.
4. **Implement Account Lockout Policies (High):** Implement account lockout policies to prevent brute-force attacks. Lock accounts after a certain number of failed login attempts.
5. **Restrict Remote Access to Database (Medium):** Restrict remote access to the database server to only authorized IP addresses. Use a VPN or other secure tunnel for remote access.
6. **Secure PID File Configuration (Medium):** Review and correct the insecure PID file configuration. Ensure that the PID file is owned by the MySQL user and group and has appropriate permissions.
7. **Investigate Aborted Connections and Timeout Errors (Low):** Investigate the root cause of the aborted connections and timeout errors. Check for network latency, server overload, or resource exhaustion.

7. INCIDENT RESPONSE

1. **Containment:**

- * Immediately block the identified malicious IP addresses (193.42.33.45, 195.54.160.149, 45.146.165.37) at the firewall.
- * Isolate the affected database server from the network to prevent further data exfiltration.

2. **Eradication:**

- * Patch the SQL injection vulnerabilities in the application code.
- * Reset the 'root' user password and enforce strong password policies.
- * Review and remove any unauthorized user accounts.

3. **Recovery:**

- * Restore the database from a clean backup if data corruption is suspected.
- * Verify the integrity of the application code and configuration files.
- * Monitor the system for any signs of further compromise.

4. **Post-Incident Activity:**

- * Conduct a thorough root cause analysis to identify the vulnerabilities that allowed the attack to occur.

*** Implement the recommendations outlined in this report to prevent future attacks.**

*** Review and update the incident response plan to incorporate lessons learned from this incident.**

* Consider engaging a security consultant to perform a penetration test and vulnerability assessment.

This report provides a snapshot of the current security posture based on the provided log data. Continuous monitoring and proactive security measures are essential to maintain a strong security posture and protect against evolving threats.

3. THREAT INTELLIGENCE & IOCs

Top Source IP Addresses (by event count)

Rank	IP Address	Events	Risk Assessment
1	10.0.1.104	23	LOW
2	10.0.4.195	21	LOW
3	10.0.10.27	20	LOW
4	193.42.33.45	19	LOW
5	10.0.1.246	19	LOW
6	10.0.4.111	19	LOW
7	10.0.10.79	18	LOW
8	185.220.100.254	18	LOW
9	10.0.9.176	17	LOW
10	45.146.165.37	16	LOW

Most Targeted User Accounts

Rank	Username	Events	Priority
1	'deploy'	39	HIGH
2	'backup'	32	HIGH
3	'jane'	31	HIGH
4	'lisa'	30	HIGH
5	'sarah'	30	HIGH
6	'emma'	30	HIGH
7	'sysadmin'	29	HIGH
8	'apache'	28	HIGH
9	'nginx'	28	HIGH
10	'mysql'	27	HIGH

5. SECURITY ALERTS

Total alerts generated: **81**. Showing top 15 alerts.

Alert #1: sql_injection

Timestamp: 2026-01-15T16:19:42 | **Source:** N/A

Message: Attack detected: sql_injection (confidence: 95%)

Alert #2: sql_injection

Timestamp: 2026-01-15T16:21:56 | **Source:** N/A

Message: Attack detected: sql_injection (confidence: 95%)

Alert #3: sql_injection

Timestamp: 2026-01-15T16:32:30 | **Source:** N/A

Message: Attack detected: sql_injection (confidence: 95%)

Alert #4: sql_injection

Timestamp: 2026-01-15T16:40:15 | **Source:** N/A

Message: Attack detected: sql_injection (confidence: 95%)

Alert #5: sql_injection

Timestamp: 2026-01-15T17:03:22 | **Source:** N/A

Message: Attack detected: sql_injection (confidence: 95%)

Alert #6: sql_injection

Timestamp: 2026-01-15T17:11:13 | **Source:** N/A

Message: Attack detected: sql_injection (confidence: 95%)

Alert #7: sql_injection

Timestamp: 2026-01-15T17:23:59 | **Source:** N/A

Message: Attack detected: sql_injection (confidence: 95%)

Alert #8: sql_injection

Timestamp: 2026-01-15T17:48:04 | **Source:** N/A

Message: Attack detected: sql_injection (confidence: 95%)

Alert #9: sql_injection

Timestamp: 2026-01-15T18:07:14 | **Source:** N/A

Message: Attack detected: sql_injection (confidence: 95%)

Alert #10: sql_injection

Timestamp: 2026-01-15T18:16:54 | **Source:** N/A

Message: Attack detected: sql_injection (confidence: 95%)

Alert #11: sql_injection

Timestamp: 2026-01-15T18:37:28 | **Source:** N/A

Message: Attack detected: sql_injection (confidence: 95%)

Alert #12: sql_injection

Timestamp: 2026-01-15T18:48:07 | **Source:** N/A

Message: Attack detected: sql_injection (confidence: 95%)

Alert #13: sql_injection

Timestamp: 2026-01-15T18:52:02 | **Source:** N/A

Message: Attack detected: sql_injection (confidence: 95%)

Alert #14: sql_injection

Timestamp: 2026-01-15T19:09:42 | **Source:** N/A

Message: Attack detected: sql_injection (confidence: 95%)

Alert #15: sql_injection

Timestamp: 2026-01-15T19:13:25 | **Source:** N/A

Message: Attack detected: sql_injection (confidence: 95%)

... and **66** more alerts. See full report for complete list.

6. LOG ANALYTICS & STATISTICS

6.1 Severity Distribution

Severity Level	Event Count	Percentage
CRITICAL	0	0.0%
ERROR	277	5.5%
WARNING	610	12.2%
INFO	4113	82.3%
DEBUG	0	0.0%

6.2 Log Type Distribution

Log Type	Count
mysql_error	5000

6.3 Top Source IP Addresses

Rank	IP Address	Event Count	Activity Level
1	10.0.1.104	23	LOW
2	10.0.4.195	21	LOW
3	10.0.10.27	20	LOW
4	193.42.33.45	19	LOW
5	10.0.1.246	19	LOW
6	10.0.4.111	19	LOW
7	10.0.10.79	18	LOW
8	185.220.100.254	18	LOW
9	10.0.9.176	17	LOW
10	45.146.165.37	16	LOW

7. RECOMMENDATIONS & REMEDIATION

◆ Critical

Immediately investigate all critical and high-severity alerts. Isolate affected systems if compromise is suspected.

◆ High Priority

Review and block suspicious IP addresses with high event counts. Implement additional monitoring on targeted accounts.

◆ Medium Priority

Enable enhanced logging for critical systems. Review access controls for accounts with failed authentication attempts.

◆ Best Practices

Ensure all systems are patched and up-to-date. Implement MFA for all privileged accounts. Regular review of security logs.

◆ Monitoring

Set up real-time alerts for threshold-based detections. Create correlation rules for multi-stage attack patterns.

◆ Response

Document and test incident response procedures. Ensure backup and recovery mechanisms are functional.

8. APPENDIX

Report Metadata

Report Generated	2026-03-26 17:43:06
Log Source Type	MySQL Error
Total Lines	0
Parsed Lines	0
Failed Lines	0
ML Attacks Detected	0
Total Alerts	81
Analysis Engine	Cyber Chakshu SIEM v2.0 + AI SOC Analyst

Disclaimer: This report is generated automatically by Cyber Chakshu SIEM system. The analysis is based on log data and machine learning models. Human review is recommended for critical security decisions.